Aaryahi Ahluwalia

+91 82901 58833 | ahluwaliaaaryahi@gmail.com | LinkedIn | GitHub | Portfolio

Education

Banasthali Vidyapith **July 2024 – May 2028**
Bachelor of Technology in Computer Science - AI *Niwai, Rajasthan*

Experience

Origen **June 2026 – July 2026**
Software Engineer Intern *Abu Dhabi, UAE*

- Authored the Integration Design Document (IDD) for the PoP Smart Parks IoT platform, defining a 6-layer architecture integrating 1,400+ sensors across 6 Abu Dhabi parks.
- Designed REST API integrations, JWT-authenticated workflows, GeoJSON spatial ingestion pipelines, and dashboard reporting interfaces for city-scale IoT operations.
- Conducted requirements analysis with Chinese IoT vendors and UAE stakeholders, producing integration specifications, risk assessments, and deployment plans.

ISIT Global **May 2025 – July 2025**
AI/Cloud Implementation Intern *Dubai, UAE*

- Contributed to enterprise AI and cloud transformation projects on Microsoft Azure for Etimad Holding, focusing on automation, intelligent workflows, and operational efficiency.
- Evaluated and designed generative AI use cases including developer copilots, enterprise knowledge assistants, document intelligence, and AIOps solutions.
- Developed technical documentation, architecture artifacts, and proof-of-concept plans supporting AI adoption and cloud modernization initiatives.
- Supported AI solution design for UAE Government hazard identification and classification systems while ensuring alignment with safety, governance, and compliance standards.

Projects

ArthSetu — AI Credit Intelligence Platform | *Python, Scikit-Learn, SHAP, Supabase (Barclay's Hackathon Finalist)*

- Built a full-stack credit intelligence platform to evaluate thin-file lending in India;
- Developed a logistic regression credit-risk model (AUC 0.77) with SHAP-based explainability, secure applicant/lender dashboards, and role-based access controls using Supabase RLS.

Agastya — Physics-Informed Exoplanet Detection System | *Python, PyTorch, GPflow, FastAPI, React*

- Built an end-to-end pipeline that detects exoplanet transits in noisy Kepler/TESS light curves using signal processing, deep learning, and explainability.
- Designed a 5-stage architecture (GP/wavelet denoising, TLS/BLS transit detection, astrophysical false-positive vetting, TCN+Transformer classifier, SHAP/attention explainability) achieving (AUC-ROC 0.96) with less than 5% false positive rate.

Macro-Financial Stress Tester | *Python, XGBoost, SHAP, Streamlit, Agent-Based Modeling, LLMs*

- Architected a full-stack agent-based systemic risk simulator for Indian financial markets with Python FastAPI backend, real-time WebSocket streaming, and React frontend.
- Integrated an LLM-powered RBI Governor agent (via Groq) to simulate policy interventions across three historically-anchored crisis scenarios; designed agent reasoning over market state and policy levers.

Technical Skills

Languages: Python, Java, C, SQL, HTML/CSS, TypeScript
AI / Machine Learning: XGBoost, Random Forest, SHAP, SMOTE, Pandas, NumPy, scikit-learn, LLMs (Llama 3/Ollama), MediaPipe
Development: React 18, Vite, Flask, Streamlit, Supabase, PostgreSQL, Docker, REST APIs, JWT Authentication
Systems & Engineering: Linux, MATLAB, Computer Vision, Image Processing, Git, Mermaid, PlantUML.
Cloud & Tools: Microsoft Azure (AZ-900), Git Hub, VS Code, AutoCAD, SolidWorks

Certifications & Soft Skills

- **Certifications:** Azure Fundamentals (AZ-900)
- **Soft skills:** • Problem Solving • Analytical Thinking • Data-Driven Decision Making • Research & Analysis • Project Planning • Presentation & Pitching • Logical Reasoning • Adaptability • Cross-Functional Coordination • Teamwork